

The Cellular Universe - Registered Predictions (v3.17)

M. Jambor; adversarial review and computations: Claude (Anthropic). Date of record: 7 July 2026. Working point: $\delta = 1/(\langle k \rangle + g_B) = 0.02297$ (derived), $\lambda = 0.10-0.15$ (only fitted parameter), $\Delta n_{\text{eff}} = 0.0535$ (derived). Data state at registration: DESI DR2 ($w_0 = -0.75 \pm 0.06$, $w_a = -0.86 \pm 0.25$); KiDS-Legacy $S_8 = 0.815$ (+0.016/-0.021).

#	Observable	Prediction	Standard / current data	Decided by (when)	Kill condition
1	N_{eff}	3.09-3.10	3.045 (inflation erases relic)	CMB-S4 / Simons (~2029-33)	-
2	n_s	0.9656 $\pm$ 0.0016 (derived)	fitted (~0.965); Planck 0.9649 $\pm$ 0.0042	CMB-S4 (sigma~0.002)	$ n_s - 0.9656 > 0.004$
3	primordial r	< 10⁻¹⁰ (no B-modes, ever)	many models hope 10 ⁻³ -10 ⁻²	LiteBIRD (~2032+)	detection r $\geq$ 10⁻³ kills model
4	H_0	66.4 km/s/Mpc	67.4 (CMB) vs ~73 (local)	ladder systematics; S4	local $H_0 \geq 72$ w/o systematics
5	S_8	0.86-0.87	KiDS-Legacy: 0.815(+0.016/-0.021)	Euclid / LSST (2026-30)	$S_8 \leq 0.78$ AND $w_a \leq -0.6$
6	w_0, w_a	-0.92..-0.94, -0.39..-0.61	-1, 0	DESI final (2026-27)	(see row 5)
7	DM direct detection	permanently null (gravity-only ash)	WIMP signals expected by many	LZ / XENONnT / DARWIN	confirmed non-grav. DM interaction
8	consistency relation	$n_s - 1 = -(3/2)\delta$, same δ shapes $w(z)$	no such link in LCDM	joint CMB x DESI x lensing fit	joint fit rejects common δ
9	Lorentz violation	none measurable ($(l_p/\lambda)^2 \sim 10^{-58}$)	exact postulate	SME, GRB dispersion	any confirmed LV
10	graviton background	thermal 0.90 K, peak 53 GHz	absent	far-future GW tech	-

Derivation chain (zero fitted parameters except λ): Poisson-Voronoi $\langle k \rangle = 48\pi^2/35 + 2 = 15.54$; capacity $C = g_B = 28$ boson carrier states; $\delta = 1/(\langle k \rangle + C)$; fuel era $w = -1 + \delta$ gives quasi-de Sitter $\epsilon = (3/2)\delta$; saturated holographic V-layer (area law measured, $p = 1.97$) gives $n_s - 1 = -\epsilon$; thermal graviton relic of Planck-rate genesis gives $\Delta n_{\text{eff}} = (8/7)(10.75/106.75)^{4/3} = 0.0535$. Machine-readable version: predikcie_cellular_universe_v3_17.csv. Reproducible pipelines accompany this record.